

TIANREN CHEN

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EDUCATION

University of California, Davis
B.S. in Computer Science (CS)

June 2025
Davis, CA

RELEVANT COURSEWORK

Courses: Machine Learning, Deep Learning, Algorithm Design, Computer Architecture, Software Engineering, Data Structures, Database Systems

RELEVANT SKILLS

Technical Skills:

- **Languages:** Python, C/C++, Java, Assembly, SQL, JavaScript, HTML/CSS
- **Frameworks/Tools:** React, Node.js, Express, Docker, GitHub, MySQL, SQLite, VS Code, PyCharm
- **Areas:** Machine Learning, Deep Learning, Data Analytics, Full-Stack Development

PROJECTS & EXPERIENCE

Predicting Movie Ratings Based on Metadata

Davis, CA

- Developed an end-to-end machine learning pipeline using integrated datasets from IMDb, Kaggle, and GitHub to predict IMDb movie ratings.
- Compared multiple predictive methods including Random Forest, XGBoost, and Artificial Neural Networks (ANN) to identify the best-performing model.
- Optimized hyperparameters and training configurations, employing advanced techniques to minimize loss and enhance model accuracy.

Text Classification and Generation with RNN Models

Davis, CA

- Designed and implemented RNN-based models for text classification and story generation using two curated datasets.
- Trained RNN models to classify textual content and generate coherent sequences from given prompts.
- Compared the performance of RNN, LSTM, and GRU architectures with various configuration changes.
- Evaluated model accuracy using learning curves and qualitative comparisons with training data.

Image Classification for Object Recognition

Davis, CA

- Built CNN models to classify hand-written digits, grayscale face images, and colored objects across three datasets.
- Trained and evaluated multiple CNN architectures, generating learning curves and performance reports.
- Experimented with kernel size, stride, padding, pooling, and loss functions to analyze their impact on model accuracy.

Graph Embedding and Node Classification with GNN Models

Davis, CA

- Developed a Graph Convolutional Network (GCN) for node classification using Cora, Pubmed, and Citeseer datasets.
- Implemented a custom GCN architecture for graph embedding and trained models on citation networks.
- Evaluated performance across datasets by analyzing learning curves and classification accuracy.
- Compared generalization across different graph structures and documented findings in a formal technical report.

Web Application with OAuth and DB Management

Davis, CA

- Built a full-stack web application using Node.js, Express, SQLite, and Handlebars.
- Integrated Google OAuth 2.0 authentication using the Passport.js library.

- Developed custom APIs and utilized fetch to retrieve emoji data from third-party services.

Distributed File System Project

Davis, CA

- Developed a multi-threaded distributed file system in C++ to manage concurrent file operations.
- Applied mutexes and condition variables to ensure thread-safe concurrency.
- Implemented RESTful APIs supporting GET, PUT, and DELETE operations.
- Used FIFO scheduling and transactional logic to maintain consistency and fault tolerance.

COLLEGE ACTIVITIES

DUBIS,
Head of Publicity and Project Management

Davis, CA
Feb 2024 – Present

- Led a 20-person team to organize 3 large public dubbing performances for 200+ attendees.
- Developed management strategies for team coordination.

WORK EXPERIENCE

American Eagle Immigration Law Firm., Software Developer

July 2025 - Present

- Collaborate with a team to develop AI tools for legal document automation.
- Co-built Auto Form, an automated immigration form system.
- Maintain databases and optimize workflow efficiency.

Golden Fastrack Corp., Summer Intern

June, 2023 - Sept 2023

- Performed software updates and maintenance.
- Managed databases for performance and integrity.
- Helped troubleshoot and optimize workflows.